CellScope AI

Session-aware quality support for organ-on-a-chip bright-field microscopy

Technical report | Review draft | 2 October 2026

Team lead: Mitja Bohak. **Proposed competition category:** End-to-End System. Public interactive demo: https://cellscope.mitja.si/. The public code repository and demo-video links are pending final release checks.

Abstract

CellScope AI is a local research prototype for prioritizing review of organ-on-a-chip microscopy images. It predicts the public dataset's binary good/bad sample-quality label from bright-field images. The model combines four global DINOv2 representations with three weakly supervised local-patch representations. Across two acquisition-session-grouped five-fold partitions of the **same** 3,072 images, balanced accuracy was **81.03%** and **79.81%**. It incorrectly called **251** and **265** source-labelled bad images good. These estimates are exploratory: the cohort was reused while choosing the architecture and no new-session independent validation exists. A broad documented search found post-hoc higher scores but no independently confirmed replacement. A training-session-only test did not confirm an optical add-on. The application supports human assessment; it does not diagnose disease or establish the biological reason for a quality label.

Evidence	Result	Meaning
First grouped partition	81.03% BA	Research development estimate
Second grouped partition	79.81% BA	Same 3,072 images, new session grouping
Full-data app model	No independent score	Refit on all development images

1. Problem and intended use

Repeated bright-field images allow researchers to monitor organ-on-a-chip cultures. A reviewer must decide which images are suitable for evaluating culture quality. The source dataset provides a binary good/bad sample-quality label; it does not provide a complete, verified per-image taxonomy of technical versus biological reasons for rejection. Our objective is to support prioritization and traceable review, not replace a biologist's judgment.

A local interface accepts one image or up to 20 images from one acquisition session. It returns a good/bad estimate and a probability-like model score. An optional experimental relative-texture warning may ask for additional review when the user supplies a sufficiently large same-session batch and cell type; it never changes the primary model decision. A separate opt-in queue lets reviewers store new images and record provisional or expert-confirmed labels.

2. Data and integrity

We use the Organ-on-a-Chip Image Dataset [1, 2]: 3,072 automated bright-field microscopy images across six cell types (A549, CACO, HPMEC, HSAEC, HUVEC, NHBE), with good/bad sample-quality labels. The local development manifest groups them into 59 acquisition sessions. Some experimental metadata fields are incomplete; the final DINOv2 candidate uses the image rather than relying on those fields.

The published train/validation/test folders contain images from overlapping acquisition sessions. Our integrity audit found 98 of 101 cell-type/session groups crossing the published folder boundary. We therefore use whole-session holdout for the main development comparisons. Images from the same acquisition session never appear in both the training and outer evaluation side of a fold.

The source data are attributed to Movcana and colleagues [1, 2]. This compact reviewer release does not redistribute the original 6.7 GB image archive. Exact terms for the relevant dataset release and sample-image reuse should be checked before public distribution.

Label scope: 'bad' can include technical imaging artifacts that impair assessment, but we cannot infer the cause of a particular image's label solely from its binary class. A coloured digital transform is also not a measured fluorescence channel.

3. Model and software

The frozen visual backbone is Meta DINOv2 [3]. We extract global representations with S/14 at 224 pixels; B/14 at 224 and 336 pixels; and B/14 after a conservative geometric channel crop at 224 pixels. Five local crops per image feed three weakly supervised local branches. Each global and local branch standardizes features and applies PCA with 96 components before a selected shallow classifier. Global components use an SVM or logistic regression; local patch components use logistic regression and a selected pooling rule. Local crops inherit the image label, not independently verified patch labels.

Stage	Implementation	Interpretation
Input	One image or a same-session batch	No extra optical channels are inferred
Frozen features	Four global and three local DINOv2 views	Pretrained visual features, not a biological foundation model
Shallow heads	Scaler + PCA + selected classifier	Training is confined to session-grouped folds
Combination	80% global + 20% local	Within-session global smoothing for batches
Output	P(good), good/bad, optional review warning	Human review remains necessary

The full-data shallow head used by the demo is refitted on all 3,072 development images. Its inference code is self-contained and does not import research-training scripts. Our replay test matched saved held-out probabilities within 0.000081. The replay checks packaging, not generalization. A single image receives no within-session smoothing and therefore cannot inherit a batch-based score.

The current application displays scores and a review queue, but does not claim verified rejection causes, disease labels, cell segmentation, or live retraining from uploads.

4. Evaluation and findings

The principal endpoint is balanced accuracy (mean of sensitivity for source-labelled bad and good classes), with threshold 0.5. We report ROC-AUC and, separately, the critical count of bad images predicted good. The two five-fold partitions use different groupings of the same 59 sessions. They are sensitivity analyses, not independent replications. Inner training sessions select model components; outer session labels score the final fold.

Candidate	Partition 1 BA	Partition 2 BA	Bad -> good
DINOv2 frozen ensemble	81.03%	79.81%	251 / 265
HPMEC optical blend, post-hoc	81.25%	80.00%	249 / 260
HPMEC optical, inner activation	81.09%	79.80%	251 / 267

The frozen ensemble has ROC-AUC 0.882 and 0.868 across the two partitions. With 1,345 bad and 1,727 good source labels, the first partition has 251 bad-as-good and 333 good-as-bad errors; the second has 265 and 357. The higher balanced accuracy should not be described as 81% safe automation.

Error at threshold 0.5	Partition A	Partition B
Bad image accepted as good	251 / 1,345 (18.7%)	265 / 1,345 (19.7%)
Good image sent to bad/review	333 / 1,727 (19.3%)	357 / 1,727 (20.7%)
All wrong binary decisions	584 / 3,072	622 / 3,072

Earlier V6 achieved 72.32% balanced accuracy with context and 72.38% from images alone under its nested grouped protocol. Because subsequent experiments repeatedly used this cohort to design the DINOv2 system, this comparison does not establish a prospectively confirmed improvement.

The HPMEC-only optical blend looked slightly better after the team inspected subgroup results. We then tested a training-session-only activation rule: include the 10% optical branch only if it strictly improves HPMEC balanced accuracy on inner calibration sessions without increasing bad-as-good errors there. It activated in 3/5 and 2/5 outer folds. The second partition lost 0.02 percentage points and added two bad-as-good errors. We therefore did not promote the optical branch. This control still uses the repeatedly studied cohort.

5. Error analysis and limitations

A review of persistent false acceptance, matched same-session pairs, optical measurements, texture, channel crops, segmentation pilots, and local student models did not establish one reliable, general visual cause for all errors. Some source-labelled good and bad images are visually similar. Our non-expert visual review cannot overrule the source laboratory's label or identify an unseen functional property of a culture.

• **Selection bias:** the same 3,072-image cohort guided many architecture decisions. Both grouped results are exploratory.

• **Limited label detail:** no verified per-image reason code separates blur, bubbles, channel damage, or biological state.

• **Shift:** a new microscope, illumination setting, cell protocol, or laboratory may alter image appearance.

• **Batch sensitivity:** session smoothing and relative texture depend on the other images submitted together.

• **Operational safety:** the model's probability is not a calibrated biological risk or an autonomous acceptance rule.

In a small targeted blind comparison of 12 same-session A549/HPMEC good/bad pairs, a non-biologist reviewer could not justify a visual choice in 10 pairs. This illustrates ambiguity in selected hard cases; it is not a measurement of biologist accuracy. The source did not publish verified per-image causes, so even visually conspicuous bubbles or channel borders cannot be treated as ground-truth explanations.

For a decisive next test, freeze the model before collecting new acquisition sessions. Ask experts to record the binary label plus separate technical-image and biological reasons, and measure inter-rater disagreement on a subset. Evaluate balanced accuracy, bad-as-good rate, calibration, and the fraction of images requiring review on those new sessions. No such independent test is claimed here.

6. Research search and negative findings

Our dated method inventory and experiment register [5] record the implemented comparisons. We tested optical and texture measurements; experimental context; EfficientNet; frozen, adapted and self-supervised DINOv2; global and local crops; Hough channel geometry; Cellpose segmentation; MorphEm; MixStyle; same-session averaging; specialist models; review thresholds; digital contrast, pseudo-colour, CLAHE and estimated flat-field. Different changes sometimes helped one cell type or partition while hurting another. Digital recolouring is not a physical fluorescence measurement.

Family	Finding	Decision
Local DINOv2 views	80.63/78.55% to 81.03/79.81% BA	Keep as candidate
Session smoothing	80.23/79.53% to 81.03/79.81% BA	Keep for batches
CLAHE / estimated flat-field	No full-model gain on both partitions	Retain raw image
HPMEC optical add-on	Post-hoc 81.25/80.00%; nested 81.09/79.80%	Do not promote
Other models and mixes	Complementary errors; uncertain transfer	Research only

A finite sweep covered all **1,020 specified** fixed CNN/DINO blends and **2,625 specified** historical one-to-three-method combinations. It did not enumerate every conceivable model or continuous weight. The selected three-model historical mix reached **82.07/80.76%** BA and 240/259 bad-as-good errors, but was selected using the same labels on which it is scored. Its paired session-bootstrap intervals include zero and do not correct search bias. This is an exploratory lead, not a validated superior model [6].

Among the implemented candidates, no more complex option has yet shown a sufficiently independent and stable improvement to replace the presented four-global/three-local model. The present score is not an information ceiling of the images.

7. How to gain new information

The next useful improvement should target missing evidence, not only new blending weights. A pilot of roughly 20-30 new independent experiments or acquisition sessions is a starting plan, not a promised accuracy gain. Record experiment, chip, session and field identifiers; capture several positions per chip while preserving raw bit depth and acquisition settings. Prioritize ambiguous A549/HPMEC cases while covering all six cell types and both quality classes. Hold out complete chips or experiments, ideally including a new device or laboratory.

New evidence	Purpose	Check on locked test
New chips and sessions	Independent morphology and acquisition variation	BA, bad-as-good, per-cell type
Two expert reviews	Separate technical and biological reasons	Agreement and reason accuracy
Matched blank + dark frames	Real illumination correction	Raw versus corrected, same folds
Optional multi-view capture	New focus, position or time information	Gain per extra acquisition cost

Before showing model scores, two experts should independently record technical assessability, culture appearance for cell type/day, a good/borderline/bad decision, confidence, region, and reason codes. Preserve disagreements and adjudication. This can support a future multi-output model with testable reasons rather than an invented explanation derived from a single binary label.

A genuine flat-field test needs an illuminated empty-field reference and a dark-camera frame matched to microscope, objective, light setting, exposure and gain. No verified matched blank/dark frames exist for the current 3,072 images. Our Gaussian-background and CLAHE experiments were approximations and did not stably improve the candidate [7, 8]. A true calibration can be compared with raw frames on newly held-out sessions; it cannot restore detail lost to blur or saturation.

Where feasible, paired focal planes, channel positions, time points or actual fluorescence could add physical information. Synthetic UV colouring cannot. Freeze the model, threshold and correction before opening the new test. Report balanced accuracy with chip/session-level intervals, bad-as-good rate, results by cell type, calibration and coverage versus risk under manual review. Keep the simpler model if the new method does not pass these criteria.

8. Reproducibility and release

A compact reviewer bundle includes source inference code, the 8.6 MB full-data shallow head, versioned Python requirements, SHA-256 file manifest, a synthetic CPU inference smoke test, and both held-out prediction CSVs. It excludes raw images and private reviewer uploads. The CSVs permit exact recomputation of the development balanced accuracy and false-accept counts without sharing the source image archive.

The package was extracted to a separate directory and installed into a new Python 3.12 environment. The bundled metric script and inference smoke test are the clean reproduction checks. The full 6.7 GB data download and GPU feature-extraction/retraining pipeline remain separate research steps and are not implied by that compact test.

The local model downloads its public DINOv2 backbone on first use. The core application binds to 127.0.0.1 and stores images only when users opt into the local review queue. The model head is serialized with Python joblib; reviewers should verify the supplied SHA-256 manifest before loading it.

The public, read-only inference demo is available without login at https://cellscope.mitja.si/. The local English narrated video is 3:07 and uses real application captures. The reviewer bundle passed SHA-256 manifest verification, both saved-metric replays and one synthetic-image CPU inference in an isolated directory on 2 October 2026. Public code and video URLs will be added only after release review. This PDF is a review draft, not evidence that a Kaggle submission has occurred.

References

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